BOARD ACTIVITY SHEET: JULY 2024

Science and Technology Part - II

Time: 2 Hours **Max. Marks: 40**

Note: i. *All* questions are compulsory.
ii Use of calculator is not allowed.
iii. The numbers to the right of the questions indicate full marks.
iv. In case of MCQs [Q. No. 1 (A)] only the first attempt will be evaluated and will be given credit.
v. For every MCQ, the correct alternative (A), (B), (C) or (D) with subquestion number is to be written as an answer.
For e.g. i. (A) ii. (B) iii. (C)
vi Scientifically correct, labelled diagrams should be drawn wherever necessary.

Q.1. (A) Choose *correct* option and write its number for the following questions: **[5]**

i. Evolution means _____ . 1
(A) Internal change (B) Gradual change
(C) External change (D) Quick change

ii. One of the rare species is _____ . 1
(A) Lesser florican (B) Red panda
(C) Tiger (D) Shekhru

iii. The machine which converts the kinetic energy of wind to electrical energy is called _____ . 1
(A) Solar cell (B) Thermal
(C) Generator (D) Wind turbine

iv. Chemically, vinegar is _____ . 1
(A) Acetic acid (B) Amino acid
(C) Hydrochloric acid (D) Gluconic acid

v. We get _____ kcal energy per gram of proteins. 1
(A) 4 (B) 3 (C) 2 (D) 1

(B) Solve the following questions: **[5]**

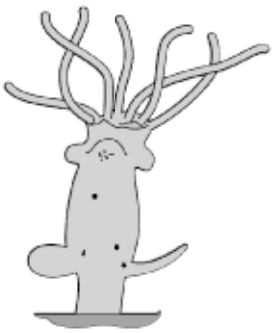
i. Find odd man out: 1
Starfish, Snail, Frog, Amoeba

ii. Match the correct pairs: 1

	Column 'A'		Column 'B'
(1)	Biotic components	(a)	Soil
		(b)	Plant
		(c)	Sunlight

iii. Write the correlation: 1
White revolution: Dairy :: Blue revolution: _____

iv. Observe the diagram and identify the phylum: 1



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v. State, whether given statement is right or wrong: 1
Peer group influence is stronger in case of adolescents.

Q.2. (A) Give scientific reasons (any *two*):

i. Fibers are one of the important nutrients.

ii. Though tortoise lives on land as well as in water, it cannot be included in class-Amphibia.

iii. Hydroelectric energy, solar energy and wind energy are called renewable energies.

(B) Answer the following questions (any *three*): **[6]**

i. Write down *four* benefits of Biotechnology.

ii. Distinguish between Sexual reproduction and Asexual reproduction.

iii. Observe the following diagrams and complete table:

(A)

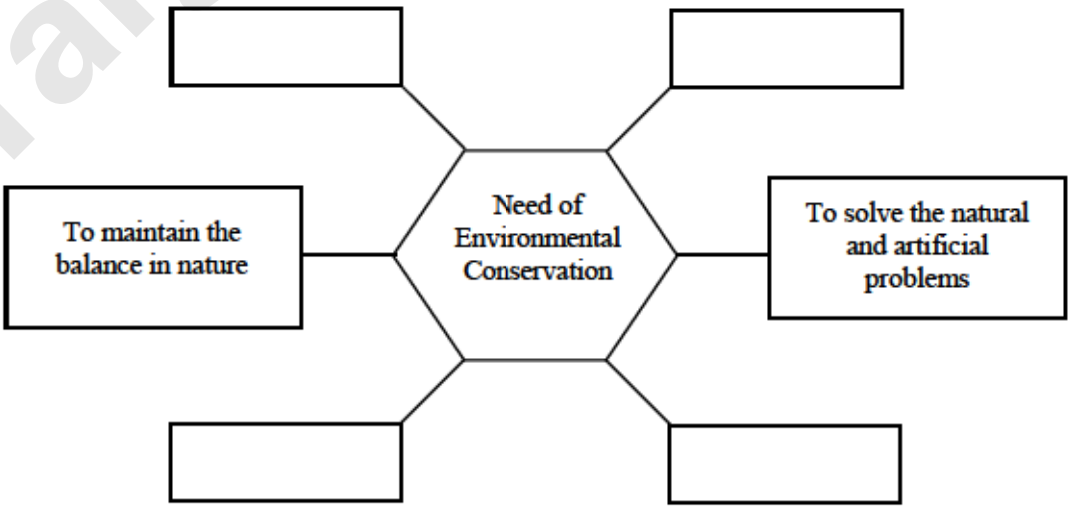
Disaster
↓
[ ]
↓
Precaution
↓
[ ]

(B)

Snake Bite
↓
Primary Stage
↓
(1)
(2)

iv. Write down any *two* features of Industrial Microbiology.

v. Complete the following flow chart:



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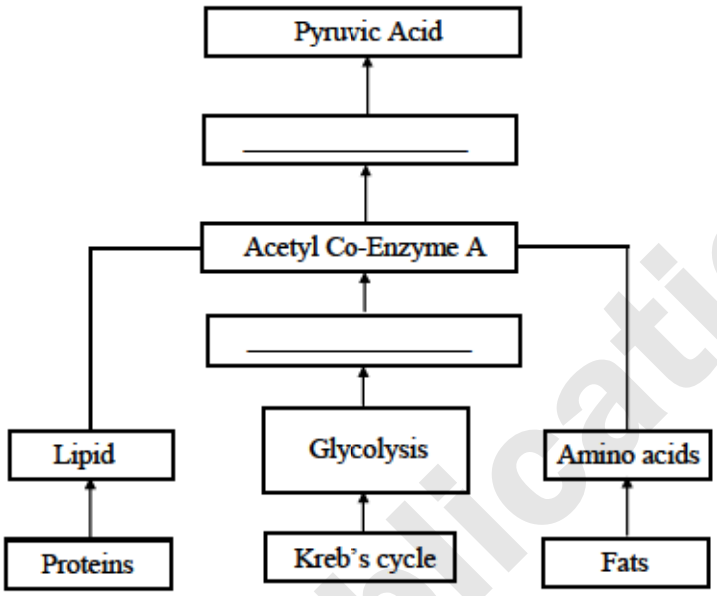
Q.3. Solve the following questions (any *five*): **[15]**

i Write any *three* characteristic features of class "Aves".

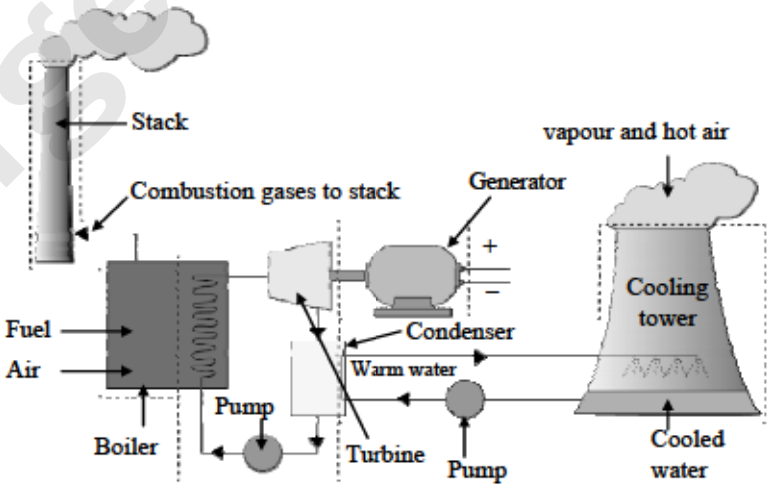
ii. (a) What are vestigial organs?
(b) Write *two* names of some vestigial organs in human body.
(c) Write the names of those animals in whom same organs are used? Give *two* examples.

iii. How energy is formed from oxidation of carbohydrates, fats and proteins?
Correct the diagram given below:



iv. What is the meaning of the following symbols? Write your role accordingly.

v. Observe the diagram and answer the following questions:



(a) Which type of energy is created in the above diagram?
(b) Which type of fuel is used in power plant?
(c) State *one* problem created in such type of power plant.

vi. Explain the importance of Biopesticides in organic farming.

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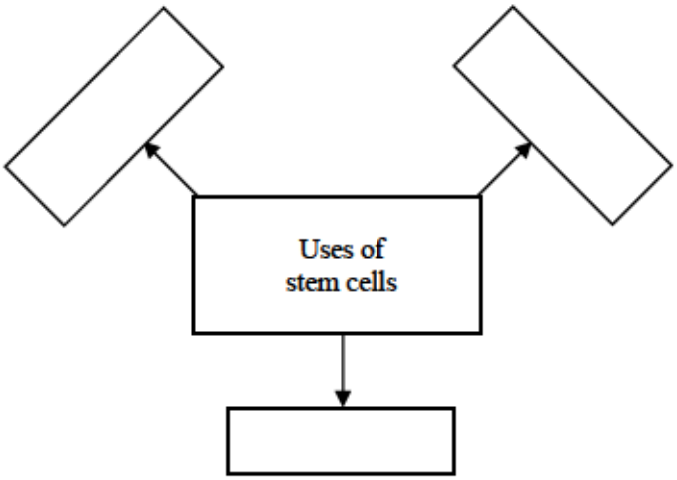
vii. Complete the following conceptual picture and write its uses in treatment of various diseases:



viii. Which are *three* objectives of disaster management?

Q.4. Solve the following questions (any *one*): **[5]**

i. (a) Draw well labelled diagram of female reproductive system. 2
(b) Which type of hormones are secreted in female reproductive system? 1
(c) When does menopause period stop? 1
(d) How many ovaries are there in female foetus at the time of birth? 1

ii. Give *two* examples of each:
(a) Hobbies to reduce stress. 1
(b) Diseases endangering the social health. 1
(c) Physical problems arising due to excessive use of mobile phones. 1
(d) Activities under the jurisdiction of cyber crime laws. 1
(e) Write problems due to alcoholism. 1

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